

UNIVERSITY OF THE FREE STATE

BLOEMFONTEIN CAMPUS

CSIS3744

MEMORANDUM

**DEPARTMENT: COMPUTER SCIENCE AND
INFORMATICS**

CONTACT NUMBER: 051 401 3702

**SICK TEST
26th November 2020**

ASSESSOR(S): 1. Prof Okuthe P. Kogeda

MODERATOR(S): 1. Mr W.S.J. Marais

TIME: 60 min

MARKS: 50

INSTRUCTIONS: Answer ALL the questions. Number your questions clearly and correctly.

Question 1 [10 Marks]

- (a) Duplicate IP address is one of the common network configuration errors. As a network specialist, how would you address this issue? [2 Marks]
*<<< Check the static settings, which might be using the wrong information; ✓
Try switching to DHCP. ✓ >>>*
- (b) What are the advantages of multilayer switching? [2 Marks]
<<< The primary advantage is that a multilayer switch allows administrators to collapse their network, creating fewer layers, while still maintaining the same level of redundancy. ✓ Provide high-speed scalability with low latency. ✓ Multi-layer switching can move traffic at wire speed and also provide layer 3 routing. There is no performance difference between forwarding at different layers

- because the routing and switching is all hardware-based – routing decisions are made by specialized ASIC with the help of content-addressable memory. ✓>>>
- (c) What are the differences between physical and logical topology? [2 Marks]
 <<< Physical topology—Mostly applies to hardware and describes how computers, other devices, and cables fit together to form the physical network ✓ while Logical topology—Has to do with software and describes how access to the network is controlled. ✓>>>
- (d) Discuss any two transmission flaws. [2 Marks]
 <<< The explanation of any two earns a candidate 2 marks.
 1. Noise - any undesirable influence degrading or distorting the signal, measured in decibels, i.e., EMI, cross-talk, etc. ✓ 2. Attenuation - loss of signal strength as it travels away from the source. ✓ 3. Latency - the delay between signal transmission and receipt, which may cause transmission errors. It is caused by cable length, intervening connectivity device, etc. ✓>>>
- (e) Discuss any two cloud computing models. [2 Marks]
 <<< IaaS: With IaaS, hardware services are provided virtually, including network infrastructure devices such as virtual servers and end user interfaces. Customers are responsible for their own application installations, data management and backup and possibly operating system. ✓ PaaS: This includes the operating system, the runtime libraries or modules the operating system provides to applications and the hardware (IaaS) on which the operating system runs. Customers are responsible for their own application installations, data management and backup. ✓ SaaS: This includes the applications, data storage and backup services as well as the platform (PaaS) on which they run. Customers only need a device to connect to the services. ✓>>>

Question 2 [24 Marks]

Table I shows the OSI 7 layer reference model, the layers numbers are provided from 1 to 7. However, the corresponding OSI layers, the hardware column and protocols column are not populated in the table. The OSI 7 layer reference model describes an idealized network communications protocol family. TCP/IP does not correspond to this model directly, as it either combines several OSI layers into a single layer, or does not use certain layers at all.

Table I: TCP/IP protocol stack

Layer number	OSI Layers	Hardware	Protocols
7	Application ^{1/2}		HTTP, ^{1/2} FTP, ^{1/2} SMTP, ^{1/2} RIP, ^{1/2} SNMP ^{1/2}
6	Presentation ^{1/2}		
5	Session ^{1/2}		
4	Transport ^{1/2}	Firewall ^{1/2}	UDP ^{1/2}
3	Network ^{1/2}	Router ^{1/2}	IP, ^{1/2} ARP, ^{1/2} ICMP ^{1/2}
2	Data link ^{1/2}	Bridge, ^{1/2} Access point ^{1/2}	PPP ^{1/2}
1	physical ^{1/2}	Hub, ^{1/2} NIC, ^{1/2} Cable ^{1/2}	

- (a) Populate the table with corresponding OSI layers. [3.5 Marks]
 <<< see completed table section >>>

- (b) Place the *firewall, router, bridge, access point, NIC, hub and Cable* hardware in their respective layers. [3.5 Marks]
 <<< see completed table section >>>
- (c) Place the *ARP, RIP, SNMP, PPP, UDP, IP, ICMP, HTTP, FTP, SMTP* protocols in their respective layers. [5 Marks]
 <<< see completed table section >>>
- (d) What is the responsibility of the data link layer of the OSI model? [2 Marks]
 <<<The data link layer is responsible for multiplexing data streams, ✓ data frame detection, ✓ medium access, ✓ and error control. ✓ It ensures reliable point-to-point and point-to-multipoint connections in a communication network. ✓ Any two >>>
- (e) Discuss the situations where UDP would be preferable to TCP. [3 Marks]
 <<< UDP would be preferable when audio & video streaming ✓, faster than TCP so would be preferred where speed is of essence ✓ and supports broadcasting & multicasting ✓, etc. >>>
- (f) Explain how TCP protocol performs a three-way handshake. [3 Marks]
 <<< The client issues a request for a connection (SYN) to the server. ✓
 The server responds with a (SYN/ACK) message to the client. ✓
 The client then responds with an (ACK) message to the server. ✓ >>>
- (g) As a network specialist, you decided to implement bus topology using UTP cable for cabling.
- (i) Explain the reasons for choosing to implement bus topology. [2 Marks]
 <<< Easy to connect a computer or peripheral to a linear bus; ✓ Requires less cable length than a star topology; ✓ >>>
- (ii) Explain reasons for choosing UTP cable. [2 Marks]
 <<< less expensive, ✓ Easier to maintain and install. ✓ >>>

Question 3 [16 Marks]

- (a) When a hacker attempts to perform a VLAN hopping attack, what are the two approaches likely to be used? [2 Marks]
 <<< The two approaches the hacker might use for VLAN hopping are as follows:
 1) Double tagging - the hacker stacks VLAN tags in Ethernet frames. When the first, legitimate tag is removed by a switch, the second, illegitimate tag is revealed, tricking a switch into forwarding the transmission on to a restricted VLAN. ✓
 2) Switch spoofing-An attacker connects to a switch and then makes the connection look to the switch as if it's a trunk line. The switch might autoconfigure its port into trunk mode when it detects trunk mode on the other end of the connection. A hacker can then feed his own VLAN traffic into that port and access VLANs throughout the network. ✓ >>>
- (b) What is a Variable Length Subnet Mask (VLSM)? [1 Mark]
 <<< Traditional subnetting reduces the waste of IP addresses, but results in multiple subnets that are all the same size. This uniformity in subnet size can be inefficient in complex networks. VLSM (Variable Length Subnet Mask) allows subnets to be further subdivided into smaller and smaller groupings until each subnet is about the same size as the necessary IP address space. This is often referred to as "subnetting a subnet." ✓ >>>
- (c) How do unmanaged switches and managed switches differ from each other? [4 Marks]

<<< An unmanaged switch provides plug-and-play simplicity with minimal configuration options and has no IP address assigned to it. ✓ Unmanaged switches are not very expensive, but their capabilities are limited and they cannot support VLANs. ✓ Managed switches, on the other hand, can be configured via a command-line interface or a web-based management GUI, and sometimes can be configured in groups. ✓ Usually, they are also assigned IP addresses for the purpose of continued management. ✓ VLANs can only be implemented through managed switches, whose ports can be partitioned into groups. >>>

- (d) Discuss two of the more common network performance metrics that are utilized? [2 Marks]

<<< Some of the more common performance metrics are as follows:

1) Utilization-This metric refers to the actual throughput used as a percentage of available bandwidth. No network should operate at maximum capacity. Identify patterns of utilization and ensure that available bandwidth accounts for utilization spikes. ✓

2) Error rate-Bits can be damaged in transit due to EMI or other interference. The calculated percentage of how often this occurs is the error rate. ✓

3) Packet drops-Packets that are damaged beyond use, arrive after their expiration, or are not allowed through an interface are dropped. Packet drops result in delayed network communications while devices wait for responses or have to resend transmissions. Knowing what's normal for your network will help you identify problems when packet drop rates vary. ✓

4) Jitter-All packets experience some latency. When successive packets experience varying amounts of latency, resulting in their arriving out of order, the user experience is degraded. This is called jitter, a problem that can be addressed through traffic management techniques. ✓>>>

- (e) Discuss the three components required to manage access control to a network and its resources? [3 Marks]

<<< The three components required are as follows:

1) Authentication - user authentication, is the process of verifying a user's credentials (typically a username and password) to grant the user access to secured resources on a system or network. In other words, authentication asks the question, "Who are you?" ✓

2) Authorization - once a user has access to the network, the authorization process determines what the user can and cannot do with network resources. In other words, authorization asks the question, "What are you allowed to do?" Authorization restrictions affect Layer 2 segmentation, Layer 3 filtering, and Layer 7 entitlements. For example, what VLAN are you assigned to? What servers or databases can you access? What commands can you run on a device? ✓

3) Accounting - the accounting system logs users' access and activities on the network. In other words, accounting asks, "What did you do?" The records that are kept in these logs are later audited, either internally or by an outside entity, to ensure compliance with existing organizational rules or external laws and requirements. ✓>>>

- (f) Why might an organization be required to undergo a security audit? [1 Marks]

<<< Certain customers-for example, a military agency-might require your company to pass an accredited security audit before they'll do business with you. Regulators require some types of companies, such as accounting firms, to host periodic security audits. But even if an audit is optional, the advantage of

having an objective third party analyze your network is that they might find risks you overlooked because of your familiarity with your environment. ✓>>>

- (g) Calculate the subnet mask that can be used to segment the 172.16.0.0 network to allow for a minimum of 6 subnets while maximizing the number of hosts per subnet? [3 Mark]

<<< 255.255.224.0 ✓>>>

End of Paper